25CEMSR10 **ASSIGNMENT-02** 14/2/26

(A.) Synthetic aperture radar (SAR) Missions and Instruments are used to Design high resolution Images of Earth's Surface using Microwave radar. SAR systems operate in airborne platform (Aircraft or Drones) and spaceborne platforms (On Satellites).

SAR Instrument is an active Microwave Sensor that-

1. Transmit radar pulses toward Earth
2. Receive reflected Signal (Backscatter)
3. Uses platform Motion to Synthesize Large antennae, produce high resolution Images.

COMPONENTS:-

- Radar Antennae
- Transmitter & receiver
- Data recorder
- Signal processor.

(B.) **Airborne SAR Mission & Instruments:-**

1. Aircraft
2. Helicopters
3. UAV (Drones)

They operate at Low Altitudes and can be deployed flexibly for specific Surveys.

Characteristics of Airborne SAR:-

- High spatial resolution
- Flexible flight planning
- Small Coverage area Compared to Satellites.
- Used for research, Military & Detailed Mapping.

Examples:-

(1) UAVSAR:- Operated by NASA.

Features:- → L Band radar
→ High precision Interferometry
→ Used for Earthquake & Deformation Studies.

Applications:-

→ Fault Monitoring
→ Landslide Mapping
→ Glacier Movement.

(2) AIRSAR:- By NASA

Features:- • Multifrequency radar (P, L & C Band)
• Polarimetric Capability

Application:- → Forest structure analysis
→ Soil Moisture Studies.

(3) **EMISAR:-** [Electromagnetic Institute SAR]
By DTU.

Features:- → Fully polarimetric SAR
→ Operate in L & C Band
→ High radiometric accuracy
→ Designed for Scientific research

Applications:- a) Soil Moisture Estimation
b) Crop Monitoring
c) LU-LC
d) Env and hydrological Studies.

(4) **E-SAR**:- (Experimental SAR):-
Developed by German Aerospace Centre (DLR)

Features:- • Multifrequency SAR (X, C, L & P)
• Quad Pol
• InSAR Support
• High resolution Imaging.

Applications:- → Forest biomass Estimation
→ Topo Mapping using InSAR
→ Glacier & Snow Monitoring
→ Geological & Terrain analysis.

(5) **Pi-SAR:-** Polarimetric & Interferometric SAR:- By JAXA (Japan aerospace Exploration agency).

Features:- → X Band SAR
→ Polarimetric, Interferometric Modes

→ Very high Spatial resolution
→ Urban & Infrastructure Studies.

<u>Applications:-</u>
- ✓ Urban Mapping
- ✓ Disaster damage assessment
- ✓ Infrastructure Monitoring
- ✓ Ground deformation Studies

(6) $\boxed{\text{F-SAR:-}}$ (<u>F-SAR airborne radar System</u>):-

<u>Features:-</u>
1. Multifrequency operation (X, C, S, L, P)
2. Fully polarimetric & Interferometric Capability.
3. Surface Monitoring
4. Calibration & Validation of Satellite SAR Missions.

<u>Applications:-</u>
- ✓ Vegetation & biomass Studies
- ✓ Soil Moisture Mapping
- ✓ Surface deformation Monitoring
- ✓ Calibration & Validation of Satellite SAR Missions.

C **Spaceborne SAR Missions** :- They are Mounted on Satellites orbiting Earth. They provide large area coverage & repeated observations.

Characteristics of Spaceborne SAR:-
→ Moderate to high resolution
→ Global or regional Coverage
→ Regular revisit Time
→ All weather Imaging Capability

1) **NISAR:-** (NASA+ISRO)

Features:- a) Dual frequency (L & S)
b) Global Monitoring
c) Soil Moisture & Deformation Mapping

Applications:-
→ Agriculture
→ Disaster Monitoring
→ Forest biomass studies.

2) **SENTINEL-1:-** Operated by ESA.

Features:- → C Band SAR
→ Short revisit Time
→ Open acess Data

Application:-
- Flood Mapping
- Land Subsidence
- Coastal Monitoring

③ **RADARSAT-2** :- operated by Canadian Space agency (CSA)

Features :- ✓ Fully polarimetric SAR
✓ High resolution Imaging.

Applications :- → Ice Monitoring
→ Maritime Surveillance.

④ **TERRA SAR-X** :- By DLR

Features :- • X Band SAR
• Very high spatial resolution

Applications :- → Urban Mapping
→ Infrastructure Monitoring

⑤ **ALOS PALSAR** :- (By JAXA)

Features :- → L Band SAR
→ Forest biomass Studies

Applications :- → Deforestation Monitoring
→ Land Deformation.

25CEM5R10 **ASSIGNMENT- 6** 14/2/26

SAR AQUISITION MODES

SAR aquisition Modes refer to different ways a SAR Collect data by changing how radar beam is steered and how long it observes Target area. Different Modes provide different Trade offs between Spatial resolution, Swath Width & Coverage area.

1. STRIPMAP MODE:-

Principle:- In StripMap Mode, radar antennae points at a fixed angle to Ground, as Satellite or aircraft Moves forward. The beam Continuously Illuminates a Strip of Terrain.

Characteristics:-

- Moderate spatial resolution
- Moderate Swath Width
- Continuous Imaging along Track.

Resolution:- 3-30m

Swath:- 30-100Km

Applications:-
✓ General LC Mapping
✓ Agriculture

Azimuth Resolution:- $\rho_a = L/2$
[L = Real antennae Length]

(2) SPOTLIGHT MODE:-

Principle:- Radar beam is Electronically steered to continuously focus on a specific area for a longer time as platform moves.

Characteristics:-
- → Spatial resolution is very high
- → Small Swath
- → Detail Imaging.

Typical resolution:- 0.5-1m

Swath:- 5-15Km

Azimuth resolution:- $\rho_a = \frac{\lambda R}{2L_{syn}}$

[λ = wavelength
R = range
L_{syn} = Extended Synthetic aperture]

Applications:-
- → Urban Mapping
- → Infrastructure Monitoring
- → Military reconnaissance
- → Detailed Terrain analysis.

(3) **SCANSAR:-**

Principle:- Radar beam rapidly switches between adjacent swaths to increase coverage area.

Characteristics:-

- ✓ Very large swath width
- ✓ Lower spatial resolution
- ✓ Faster area coverage.

Typical resolution:- (20-100)m

Swath:- 400-500Km

Azimuth resolution:- $\rho_a = \frac{L}{2N}$

[∵ N = Number of subswaths, resolution decreases as swath increases]

[Additional azimuth beam steering to maintain uniform Doppler centroid]

Applications:-

- ✓ Flood Mapping
- ✓ Sea Ice Monitoring
- ✓ Large area Disaster assessment
- ✓ Regional land Monitoring.

(4) **TOPS MODE:-** (Terrain Observation with Progressive Scans)

Principle:- In TOPS Mode, radar beam is steered in azimuth during aquisition to Ensure uniform Image Quality across Swath.

Characteristics:-

→ Wide Swath
→ Improved radiometric Quality
→ Reduced Scalloping effects.

Azimuth resolution:-

$$\rho_a = \frac{L}{2}$$ (Maintain Uniform Doppler Centroid).

Applications:-

✓ InSAR
✓ Land deformation Monitoring
✓ Earthquake Studies.

This Mode is widely used in Modern Satellite SAR Missions.

(5) SLIDING SPOTLIGHT MODE:-

Principle:- This Mode is Compromise between StripMap & Spotlight.

Characteristics:-
→ High resolution
→ Moderate Swath

Sliding spotlight Mode Azimuth resolution:-

$$\rho_a = \frac{\lambda L}{2 L_{eff}}$$

(L_{eff} is larger than StripMap but Smaller than spotlight)

Applications:-
✓ Urban & regional Mapping
✓ Precision Terrain analysis.

(6) WAVE ~~Ben~~ MODE:-

Principle:- In wave Mode, Small Isolated patches of Ocean Surface are Imaged at regular Intervals Instead of Continuous Strips.

Characteristics:-
✓ Small Image scenes
✓ Periodic aquisition
✓ Ocean observation.

Applications:-
→ Ocean wave spectrum analysis
→ Wind speed Estimation
→ Marine forecasting.

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COMPARISION TABLE

MODE	RESOLUTION	SWATH WIDTH	USE
1) Strip Map	Medium	Medium	General Mapping
2) Spotlight	Very high	Small	Detailed Imaging
3) Scan SAR	Low	Very Large	Disaster Monitoring
4) TOPS	Medium	Wide	Deformation Studies
5) Sliding Spotlight	High	Medium	Urban mapping
6) Wave Mode	Specialized	Small Scenes	Ocean studies.

(7) **Range resolution:-**

[Common to all satellite Modes]

$$\rho_r = \frac{c}{2B}$$

[c = speed of light
B = Bandwidth]

SUMMARY:-

a) Strip Map:- Resolution depends on antenna length

b) Spotlight:- Improved resolu, using longer SAR

c) Scan SAR:- Wider Swath, lower resolution

d) Range resolution:- depends on bandwidth.

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